

The boomi logo is displayed in white lowercase letters on a purple-to-orange gradient background. The background features abstract, overlapping circular shapes and wavy line patterns.

boomi

Business Rules Step Activity

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About the Activities

To complete the Activities, log into the [Boomi Enterprise Platform](#) with your Training Account, then proceed with the instructions. We recommend completing the activities in order as they build on each other.

Description

Athens Books is a company that buys and sells books and is looking to automate its order management decisions. Whenever Athens Books receives an order, it needs to check if a shipping warehouse has been assigned, a discount is included, and if all contacts or approvers have an email address. In this activity, the Boomi consultant helps Athen Books create and configure the **Business Rules** step to manage these checks.



Activity: Create a Business Rules step to automate multiple, repetitive tasks

In this activity, you will complete the following tasks:

- Create a Business Rules subfolder and install the Business Rules process from the Process Library.
- Configure the Mail Connector's Connection and Operation
- Test the process and view the results
- Configure the Business Rule step
- Add Inputs to the Business Rule
- Add Conditions to the Business Rule
- Configure the error message
- Add a second Business Rules step
- Add a Group
- Add a third Business Rules step
- Format and capture the Error Message

Create a Business Rules subfolder and install the Business Rules process from the Process Library.

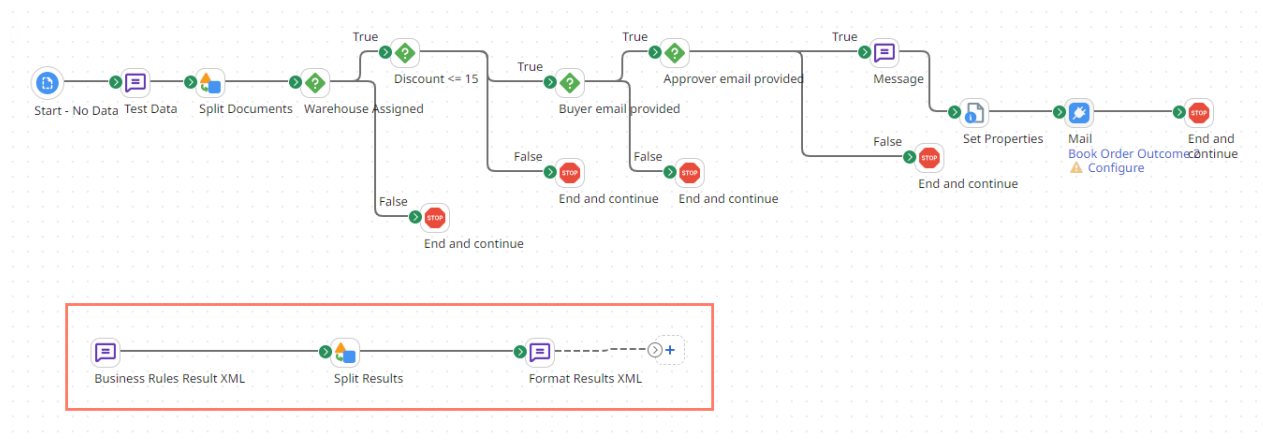
1. Create a **Business Rules** folder for your process directly from the top root folder.

Not sure how to create a folder? [Check out Create a Folder.](#)

2. Install **Business Rules Activity – START** from the **Process Library**.

Not sure how to Install a Process from the Process Library? [Check out Install a Process from the Process Library.](#)

3. Review each step of the process. The highlighted steps at the bottom contain formatting for the results message. You will connect them to the main process at the end of the Activity.



Configure the Mail Connector's Connection and Operation

1. This activity uses a reusable Connection from the Associate Developer Course. Open the Mail **Connector**, select **Browse** next to **Connection**, and select **Boomi Training (Mail)**. Select **OK to save the connector**.

Connector Shape ⁱ

Use connector shapes to get data into and out of a process. Most processes have one "get" connector and one or more "send" connectors. The Connector shape uses a combination of predefined connection and operation components to establish where and how to get or send data.

General	Parameters
Display Name	
Connector ⁱ	Mail ▼ Use the Mail connector to read email from a POP email server and send email to an SMTP email server. Learn more about this connector. ↗
Connection ⁱ	Boomi Training (Mail) ✎ ✕
Action	Send ▼
Operation ⁱ	Choose... ⊕

OK

Cancel

2. Select the **Book Order Outcome** Mail Operation.

General Parameters

Display Name

Connector 

Mail 

Use the Mail connector to read email from a POP email server and send email to an SMTP email server.
Learn more about [this connector](#). 

Connection 

Q Boom! Training (Mail)  

Action

Send 

Operation 

Q Book Order Outcome  

3. In the **To** field, enter your **email address** to replace *****ENTER YOUR EMAIL*****.
Select **Save and Close** and **OK**.

 Book Order Outcome

Mail Operation   

Save & Close

Save

Options Archiving Tracking Caching

Connector Action

Send 

From

boomitrain@gmail.com

To

*****ENTER YOUR EMAIL*****

Subject

Order Status

Disposition

Inline 

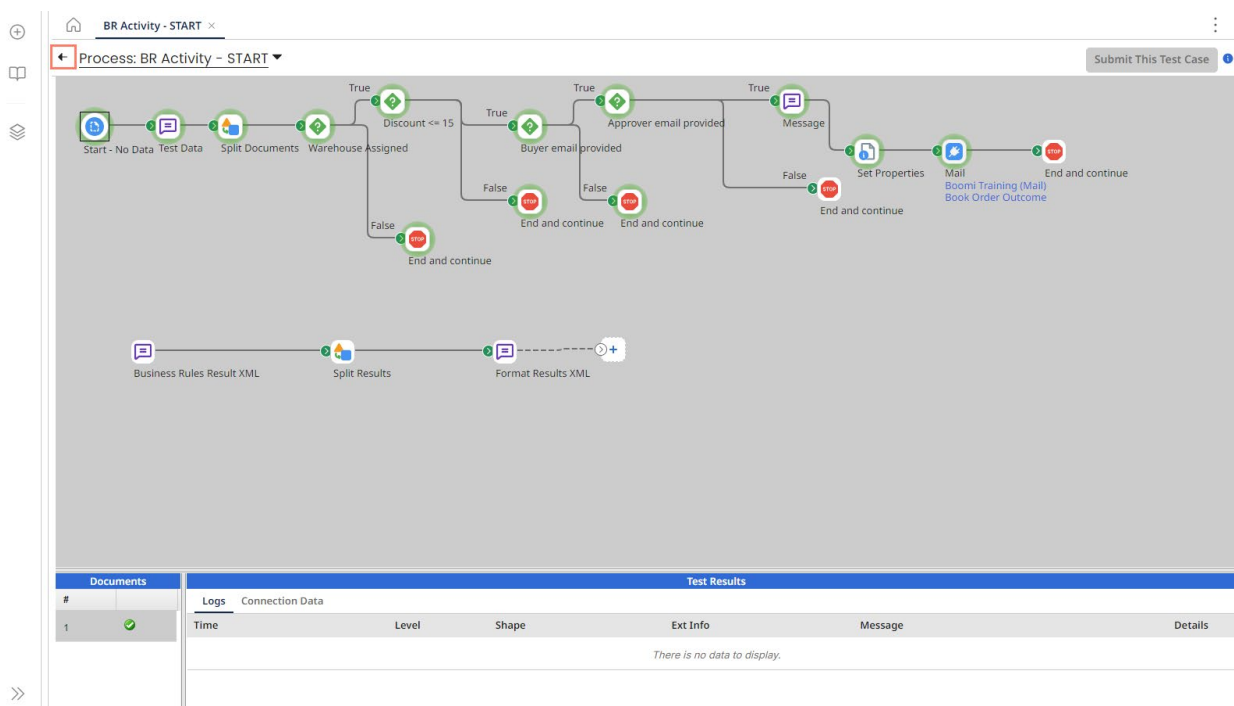
Content Type

text/plain 

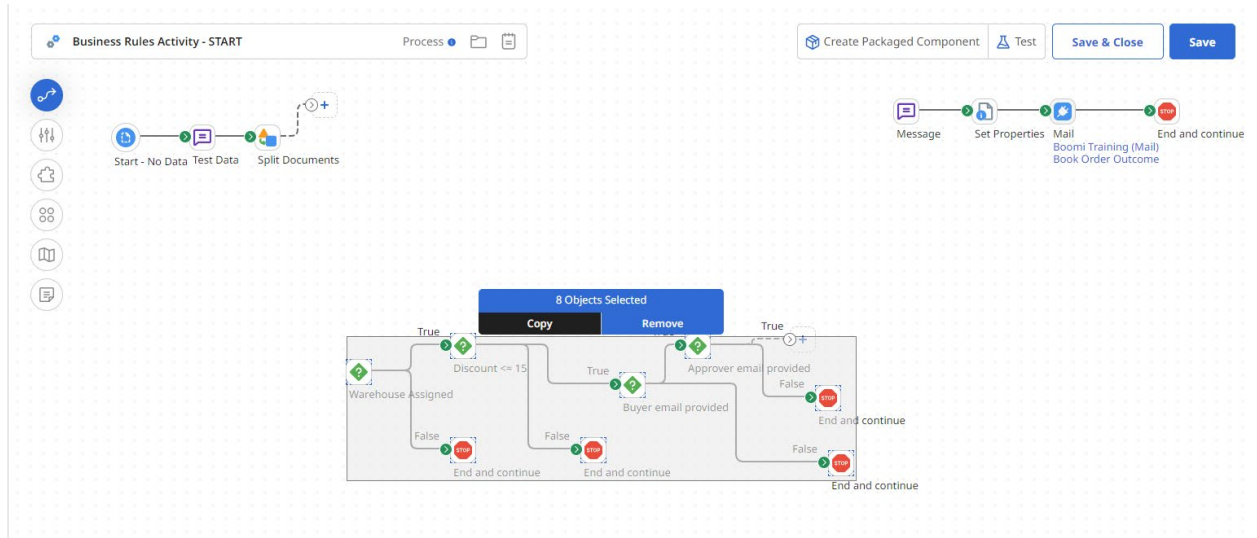
Test the process and view the results

1. Run the process on the Build page to examine the functionality and output documents. Documents will flow down several different paths. Use the **Document** viewer to examine them. Select any step that is highlighted in green or red, then look under the **Test Results** bar at the bottom of the canvas to see the logs and documents for the selected step.

Not sure how to test your process? [Check out How to Test Your Process in Test Mode.](#)



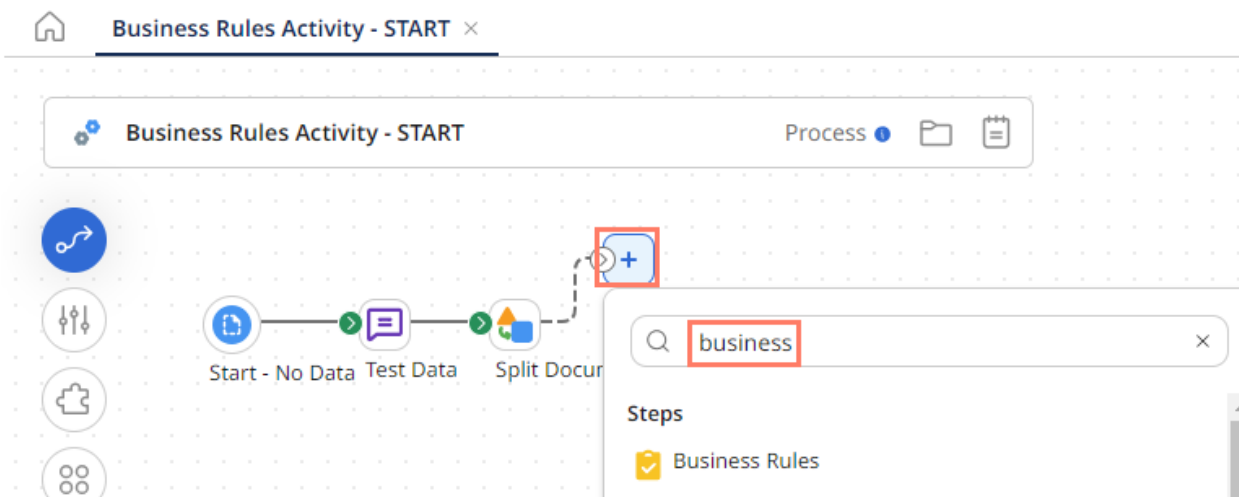
2. Select **Return to Edit Mode** arrow located above the process.
3. Detach the four Decision steps, and then highlight the steps. Select one Decision step to move them to the bottom of the canvas.



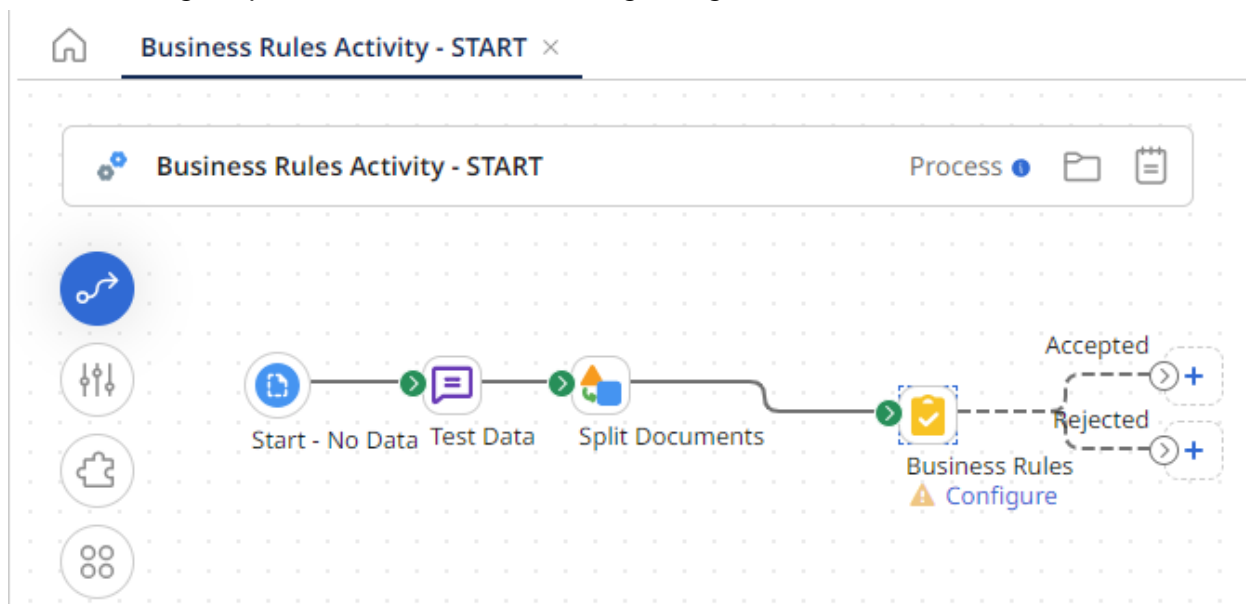
Note: Do not delete the Decisions; instead, replicate the same logic in the Business Rules step.

Configure the Business Rule step

1. Select the + icon after the Split Documents step and choose a Business Rules step.



The resulting step will look like the following image.



2. Select **Configure**.

3. Open the **Business Rules** step and enter **Book Order Profile** for the Display Name, select **XML** for the Profile type and load the provided Profile, **Book Order XML**.

Business Rules Shape Options ⁱ

To create multiple, advanced validation rules that determine if incoming documents should be accepted or rejected, use the Business Rules shape. Based on whether your documents were accepted or rejected, you can perform business logic in your process. All rules are evaluated against each document.

Display Name: Book Order Profile

Profile Type: XML

Profile (required) ⁱ Choose...

Search components

Recommended

- Book Order XML

All

- Boomi_Trainer
- Process Library

OK

4. Select **OK**.

5. Select the **Create a New Rule** button. You'll type the name for the new rule in the title bar, highlighted below.

Business Rules Shape

Rules

New Business Rule

Inputs

Add

Conditions

Top Level Operator AND OR

AND Add a Condition Add a Group

Error Message

Insert Value

OK Cancel

6. Name the Business Rule **Discount <= 15%**. Press **Enter**.

Rules

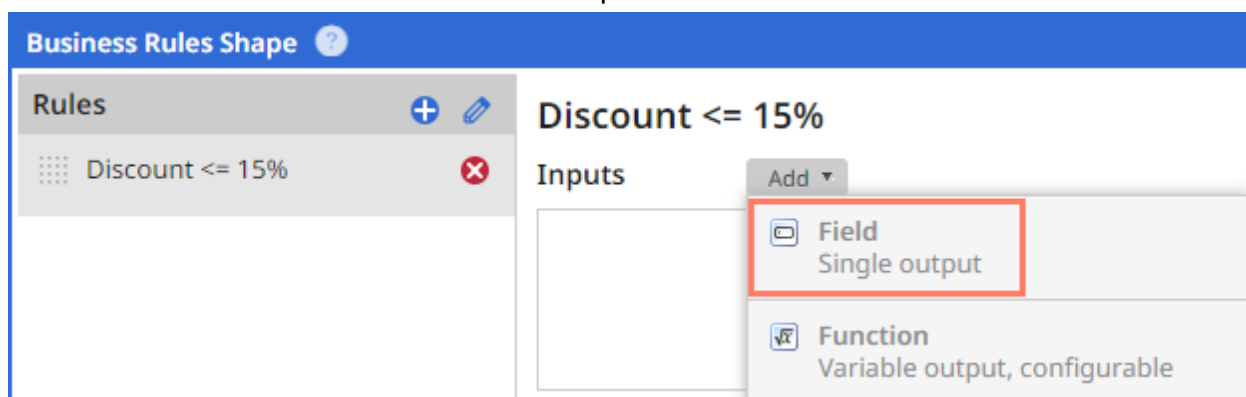
Discount <= 15%

Inputs

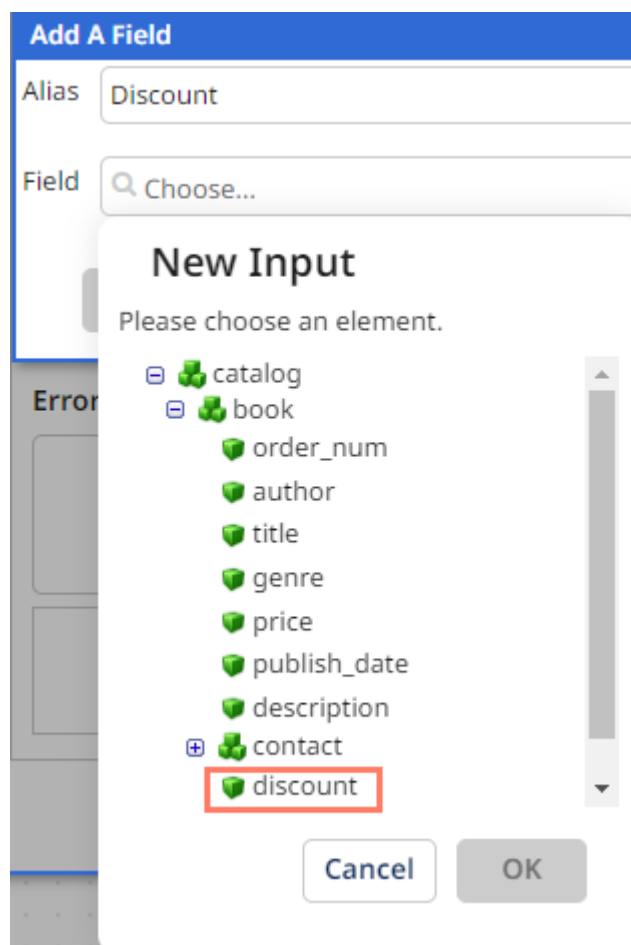
Add

Add Inputs to the Business Rules step

1. Select **Add** and select **Field** from the drop-down list.



2. Enter **Discount** as the Alias, and select the **Discount** field **OK**.



3. Repeat steps 1 and 2 of this section of the activity to add **Order** as the Alias and **order_num** as the field. You will need this field for your error message. Select **OK**.

Add A Field

Alias

Order

Field

Q

order_num (catalog/book/order_num)

OK

Cancel

Add Conditions to the Business Rules step

Follow these instructions to set the condition, Discount <= 15.

1. Select **Add a Condition**.

The screenshot shows the 'Business Rules Shape' interface. On the left, a 'Rules' panel lists 'Discount <= 15%' with a red 'X' icon. The main area is titled 'Discount <= 15%'. Under the 'Inputs' section, there are two entries: 'Discount' (discount (catalog/book/discount)) and 'Order' (order_num (catalog/book/order_num)). Under the 'Conditions' section, there is a 'Top Level Operator' set to 'AND' and a button labeled 'Add a Condition' which is highlighted with a red box.

2. Create a condition where the **Discount** field is **Less Than or Equal (<=)** option and enter a static value of **15** and select **Save**.

Discount <= 15%

Inputs

Add ▼

✖ Discount

discount (catalog/book/discount)

Conditions

Top Level Operator

AND

OR

AND

Discount ▼

<= ▼

Static ▼

Save

Cancel

15

Note: Notice the overall condition is based on conjunction, so the Top-Level Operator is **AND**.

Conditions

Top Level Operator **AND** OR

Discount <= "15"

AND [Add a Condition](#) [Add a Group](#)

Note: Added conditions appear in the Business Rules Properties pane conditions section.

Configure the Error Message

Configure a dynamic error message for each order **NOT** meeting this condition.

Note: The Error Message references dynamic parameters unique to each document analyzed by the Business Rules.

1. Use the Insert Value menu to select the fields you want to include in your error message.

Discount <= 15%

Inputs

Add ▼

✕ Discount

discount (catalog/book/discount)

✕ Order

order_num (catalog/book/order_num)

Conditions

Top Level Operator **AND** OR

Discount <= "15"

AND [Add a Condition](#) [Add a Group](#)

Error Message

Insert Value ▼

Discount

Order

- Reference each value by the number assigned by the Boomi Platform after you insert it.

Error Message Insert Value ▾

{1}

✖ 1. Order

- Enter the message "Order {1} has a discount of {2}. Please return to the approver to correct the discount." When finished with the message, select **OK**.

Business Rules Shape ⓘ

Rules + ✖

Discount <= 15% ✖

Discount <= 15%

Inputs Add ▾

✖ Discount discount (catalog/book/discount)

✖ Order order_num (catalog/book/order_num)

Conditions Top Level Operator AND OR

Discount <= "15"

AND [Add a Condition](#) [Add a Group](#)

Error Message Insert Value ▾

Order {1} has a discount of {2}. Please return to the approver to correct the discount.

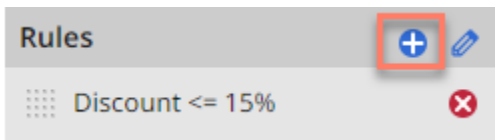
✖ 1. Order ✖ 2. Discount

OK Cancel

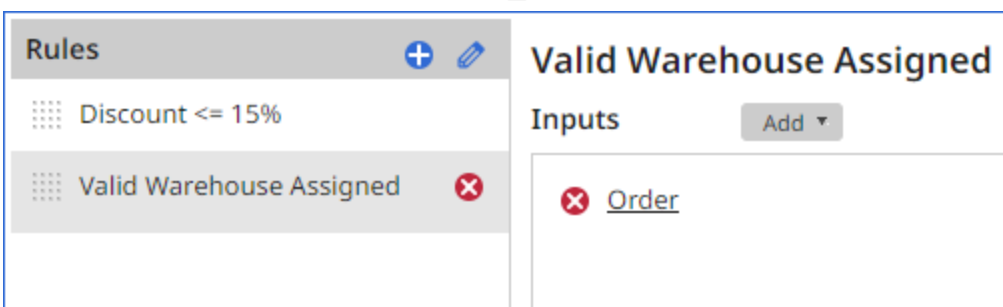
Note: An error occurs if the error message contains a hard-coded return.

Add a second Business Rule

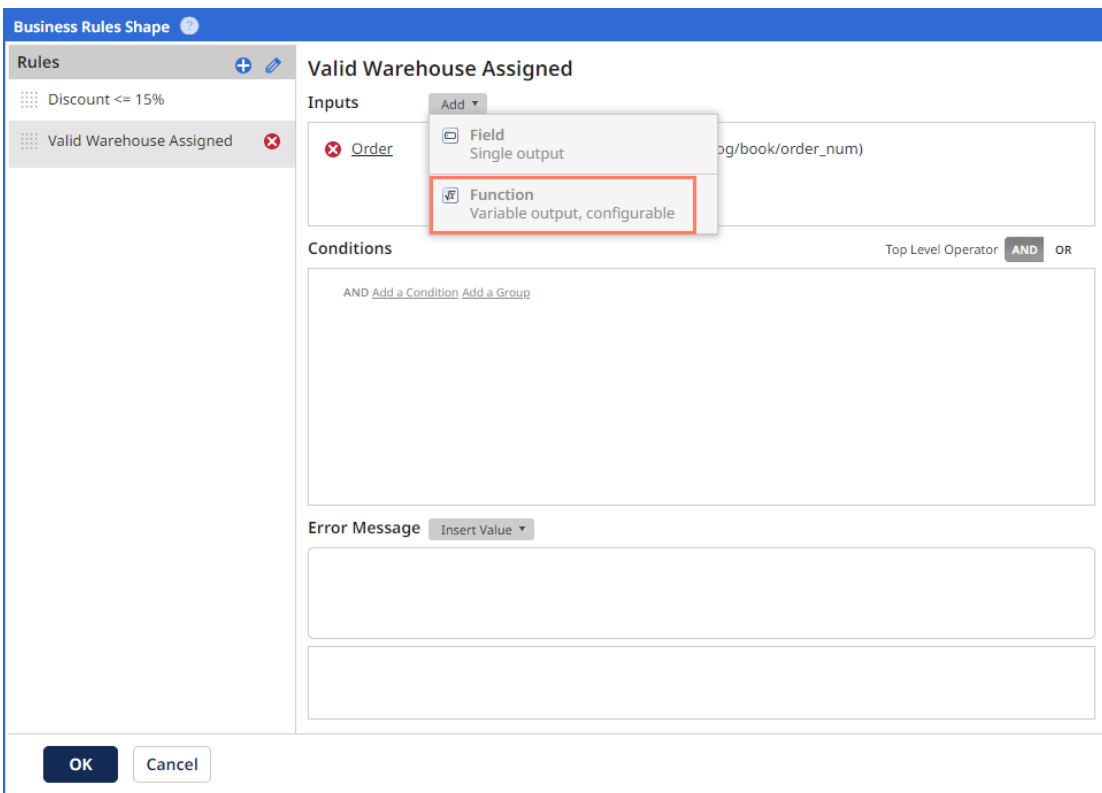
1. The second Business Rules step checks for a valid warehouse location. Add a second rule by selecting the plus icon.



2. Enter **Valid Warehouse Assigned** as the rule and press **Enter**. Select Add Field, enter **Order** as the Alias and **order_num** as the Field.



3. Next, add a **Function** to look up the Warehouse location.



4. Select the **Lookup** Category, **Cross Reference Lookup** function and **OK**

Add a Function

Add a Function

Category Lookup

Functions

- Sql Lookup
- Cross Reference Lookup
- Document Cache Lookup
- Simple Lookup

5. Select the provided Cross Reference Table, **Warehouse Location**.

Cross Reference Lookup

Cross Reference Table Choose...

Lookup References
Choose the references that will be inputs to your cross reference.

Boomi_Trainer

Process Library

Business Rules

Warehouse Location

OK Cancel

6. Select **Location** as both the Input and the Output and select **OK**.

Cross Reference Lookup

Cross Reference Table

Warehouse Location

Q ✎ +

Lookup References + ✕

Choose the references that will be the inputs to your cross reference.

Location

Values + ✕

Choose the references that will be the values, or outputs, of your cross reference.

Location

OK

Cancel

7. The next selection window allows you to associate the correct XML profile field to query the **Cross Reference Table**. Enter **Warehouse** as the Alias and select **warehouse_location** for the location. You may need to scroll to reach the field.

Add a Function

You can enter optional default values here. A default value will be i

Alias	Warehouse
Location	Choose...

New Input

Please choose an element.

- book
 - order_num
 - author
 - title
 - genre
 - price
 - publish_date
 - description
- contact
 - discount
 - warehouse_location**

Cancel OK

8. After completing the Cross Reference Lookup, select **OK**.

Add a Function

You can enter optional default values here. A default value will be used if no value is found.

Alias	Warehouse
Location	warehouse_location (catalog/boc Q

OK	Cancel
-----------------------------------	---------------------------------------

Now you have the input data you need, and you can create the conditions to test the data. Since there are multiple warehouses that you can ship books from, the recommended method is to create a group. The conditions within this group must select warehouses that exist in the table.

Add a Group

1. Select **Add a Group**.

Valid Warehouse Location

Inputs

Add ▾

✕ <u>Order</u>	order_num (catalog/book/order_num)
✕ <u>Warehouse</u>	CrossRefLookup[warehouse_location (catalog/book/warehouse_location)]

Conditions

Top Level Operator **AND** OR

AND Add a Condition **Add a Group**

2. Select **OR** as the Group Level Operator and select **Save**.

Valid Warehouse Location

Inputs

Add ▾

✕ <u>Order</u>	order_num (catalog/book/order_num)
✕ <u>Warehouse</u>	CrossRefLookup[warehouse_location (catalog/book/warehouse_location)]

Conditions

Top Level Operator **AND** OR

AND Group Level Operator

AND ▾
AND
OR

Save

Cancel

3. Within the Group, select **Add a Condition**.

Conditions

(

OR [Add a Condition](#) [Add a Group](#)

)

AND [Add a Condition](#) [Add a Group](#)

4. Set the Warehouse **equal** to a **Static** value of **New York**, and select **Save**.

Conditions

Top Level Operator **AND** OR

OR

Warehouse

▼

=

▼

Static

▼

Save

Cancel

New York

)

AND [Add a Condition](#) [Add a Group](#)

5. Create two more conditions, one for a warehouse in Chicago and the other for a warehouse in Denver.

Valid Warehouse Location

Inputs

Add ▼

✕ <u>Order</u>	order_num (catalog/book/order_num)
✕ <u>Warehouse</u>	CrossRefLookup[warehouse_location (catalog/book/warehouse_location)]

Conditions

Top Level Operator **AND** OR

```

(
  Warehouse = "New York"
  OR Warehouse = "Chicago"
  OR Warehouse = "Denver"
)
AND Add a Condition Add a Group

```

6. Create the error message, "Order {1} warehouse location {2} currently is unavailable," by inserting the values for Order and Warehouse Location.

Error Message

Insert Value ▼

Order {1} warehouse location {2} currently is unavailable. Please return to approver for correction.

✕ 1. Order ✕ 2. Warehouse

Add a third Business Rule

1. Call the rule **Contact Emails Provided**.

The screenshot shows the Business Rules configuration interface. On the left, a 'Rules' list contains three rules: 'Discount <= 15%', 'Valid Warehouse Assigned', and 'Contact Emails Provided' (highlighted with a red 'x' icon). On the right, the 'Contact Emails Provided' rule configuration is shown with an 'Inputs' section and an 'Add' button.

2. Enter **Order** as the alias and **order_num** as the Field.

The screenshot shows the 'Add A Field' dialog box. The 'Alias' field is set to 'Order'. The 'Field' dropdown is open, showing a list of elements. 'order_num' is selected and highlighted with a red box. The list includes 'catalog', 'book', 'order_num', 'author', 'title', 'genre', 'price', 'publish_date', 'description', 'contact', and 'discount'. 'Cancel' and 'OK' buttons are at the bottom.

3. Add the **Buyer** email address as a Field input, with the alias **Buyer Email**.

New Input

Please choose an element.

- author
- title
- genre
- price
- publish_date
- description
- contact
- buyer**
- approver
- discount
- warehouse_location

Cancel OK

4. Add the **Approver** email address as a Field input, with the alias **Approver Email**.

New Input

Please choose an element.

- author
- title
- genre
- price
- publish_date
- description
- contact
- buyer
- approver**
- discount
- warehouse_location

Cancel OK

5. **Add a Condition** to check that the Buyer Email is not empty. You need to scroll to find the **'is not empty'** selection.

Contact Email Provided

Inputs Add ▾

✕ <u>Order</u>	order_num (catalog/book/order_num)
✕ <u>Buyer Email</u>	buyer (catalog/book/contact/buyer)
✕ <u>Approver Email</u>	approver (catalog/book/contact/approver)

Conditions Top Level Operator **AND** OR

AND Buyer Email ▾ is not empty ▾ Save Cancel

<=
>=
contains
does not contain
starts with
is empty
is not empty

6. Add a second Condition to check that the Approver email is not empty.

Contact Email Provided

Inputs Add ▾

✕ <u>Order</u>	order_num (catalog/book/order_num)
✕ <u>Buyer Email</u>	buyer (catalog/book/contact/buyer)
✕ <u>Approver Email</u>	approver (catalog/book/contact/approver)

Conditions Top Level Operator **AND** OR

Buyer Email is not empty

AND Approver Email ▾ is not empty ▾ Save Cancel

The final rule looks like the image below:

Contact Email Provided

Inputs Add ▾

✕ Order	order_num (catalog/book/order_num)
✕ Buyer Email	buyer (catalog/book/contact/buyer)
✕ Approver Email	approver (catalog/book/contact/approver)

Conditions Top Level Operator AND OR

Buyer Email is not empty
AND Approver Email is not empty

AND [Add a Condition](#) [Add a Group](#)

- Configure the **Error Message** by adding inputs for Order, Approver Email, and Buyer Email. "Order {1} does not have emails for all contacts.
Approver email: {2}
Buyer Email: {3}"

The result looks like the image below:

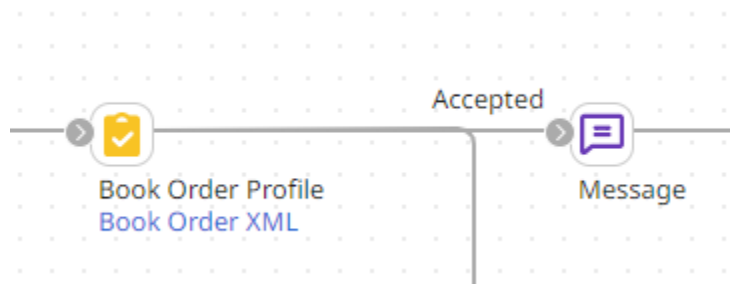
Error Message Insert Value ▾

Order {1} does not have emails for all contacts.
Approver email: {2}
Buyer email: {3}

✕ 1. Order
✕ 2. Approver
✕ 3. Buyer

- Select **OK** to close the Business Rules.

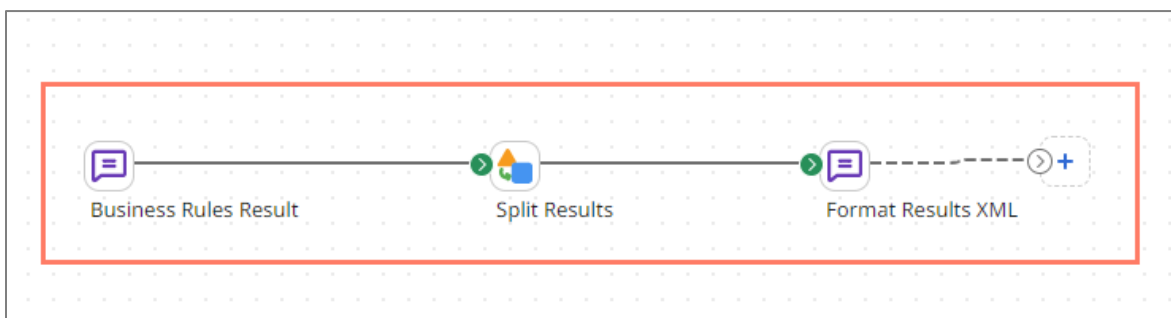
Be sure the Business Rules Step is connected to the Accepted path as shown below.



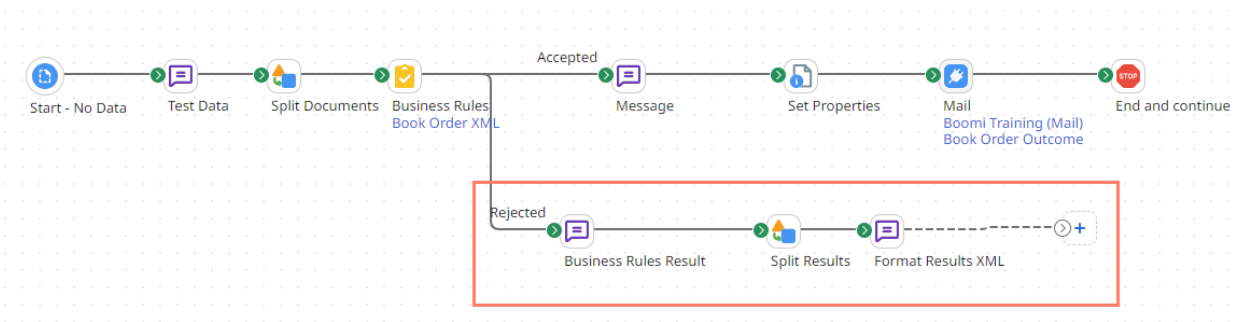
Format and capture the Error Message

Next, set up the rejected path to capture the error messages configured in the Business Rules Properties pane. This section uses the steps that were included on the canvas but not originally attached to the process.

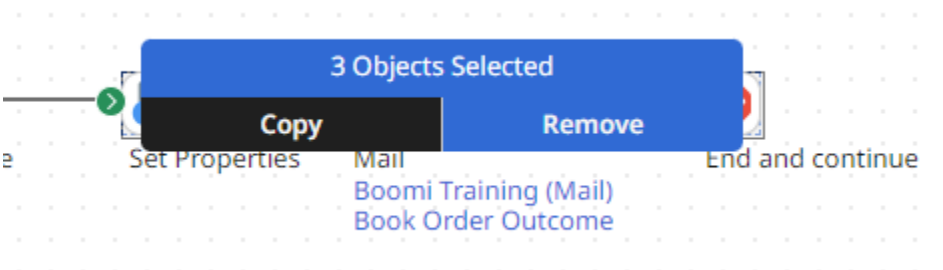
1. Find this sequence of steps on the canvas.



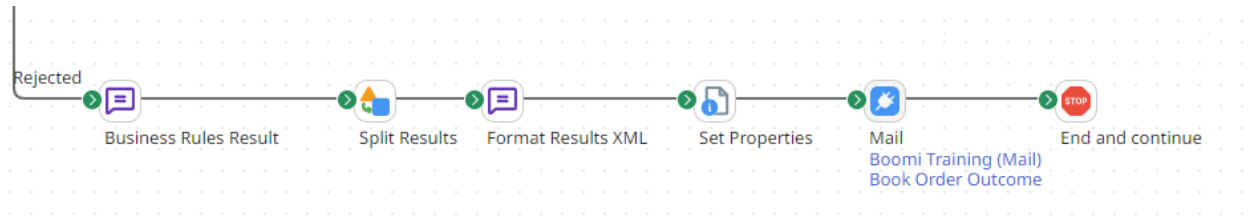
2. Attach the **Business Rule Result** step to the end of the Rejected path.



3. To complete the Business Rules process, select the last three steps in the **Accepted** path and select **Copy**.



4. Paste the selected objects at the end of the Rejected path after the **Format Results XML** message. The final rejected path should look like the following image:



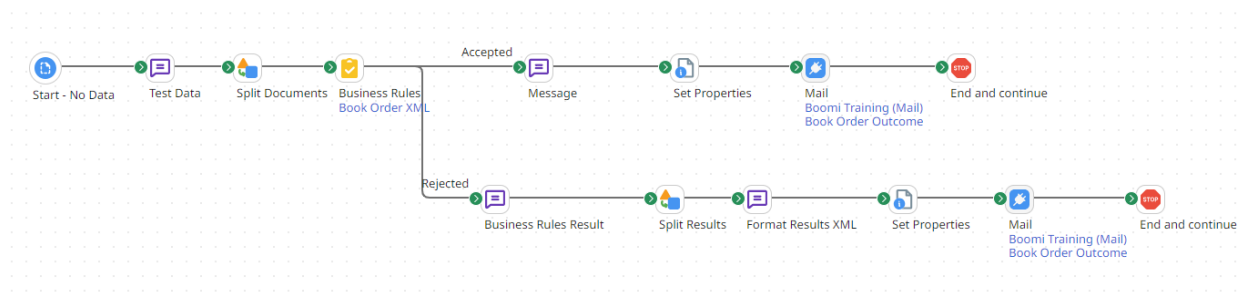
5. Update the **Set Properties** step in the Rejected path. Set the **Mail - Subject** to **Incomplete Orders as of [Date]**.

Display Name

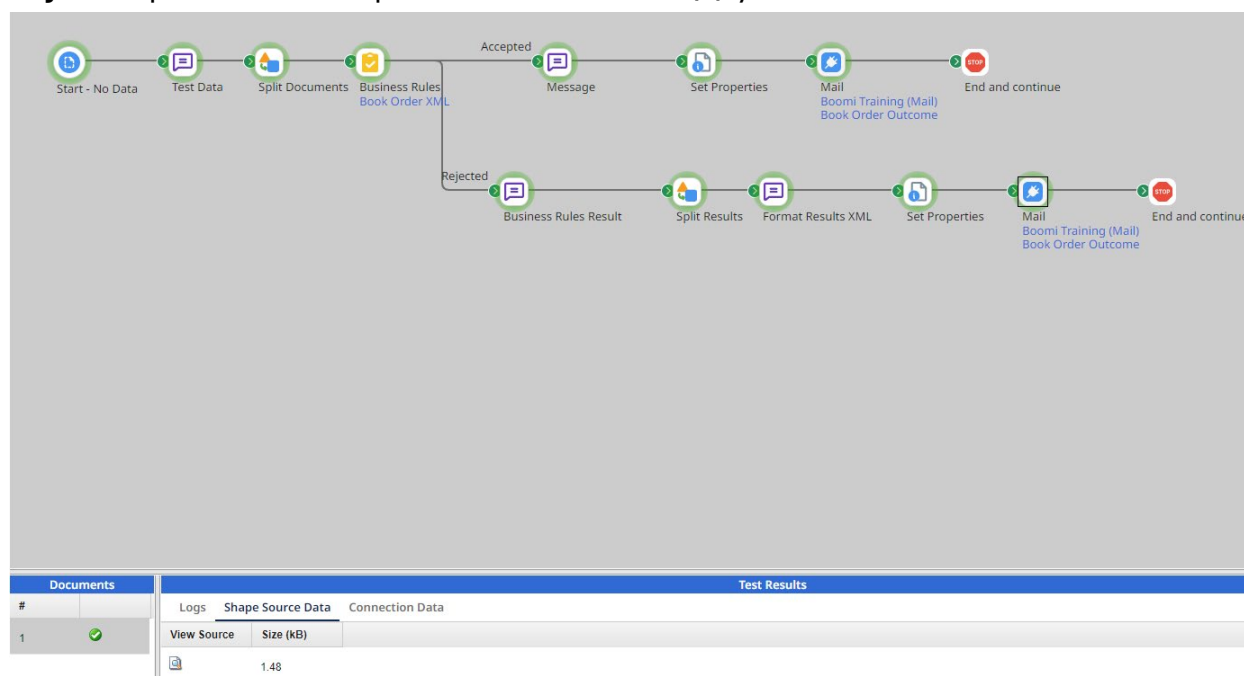
Properties to Set	Property Value (i)
+ ✎ ✖ Mail - Subject	+ ✎ ✖ ⬆ ⬇ Static value of 'Incomplete Orders as of ' Static value of " Date/Time - Current Date - MMddyyyy

Note: Do not modify the Mail Connection.

The complete process should look like this:

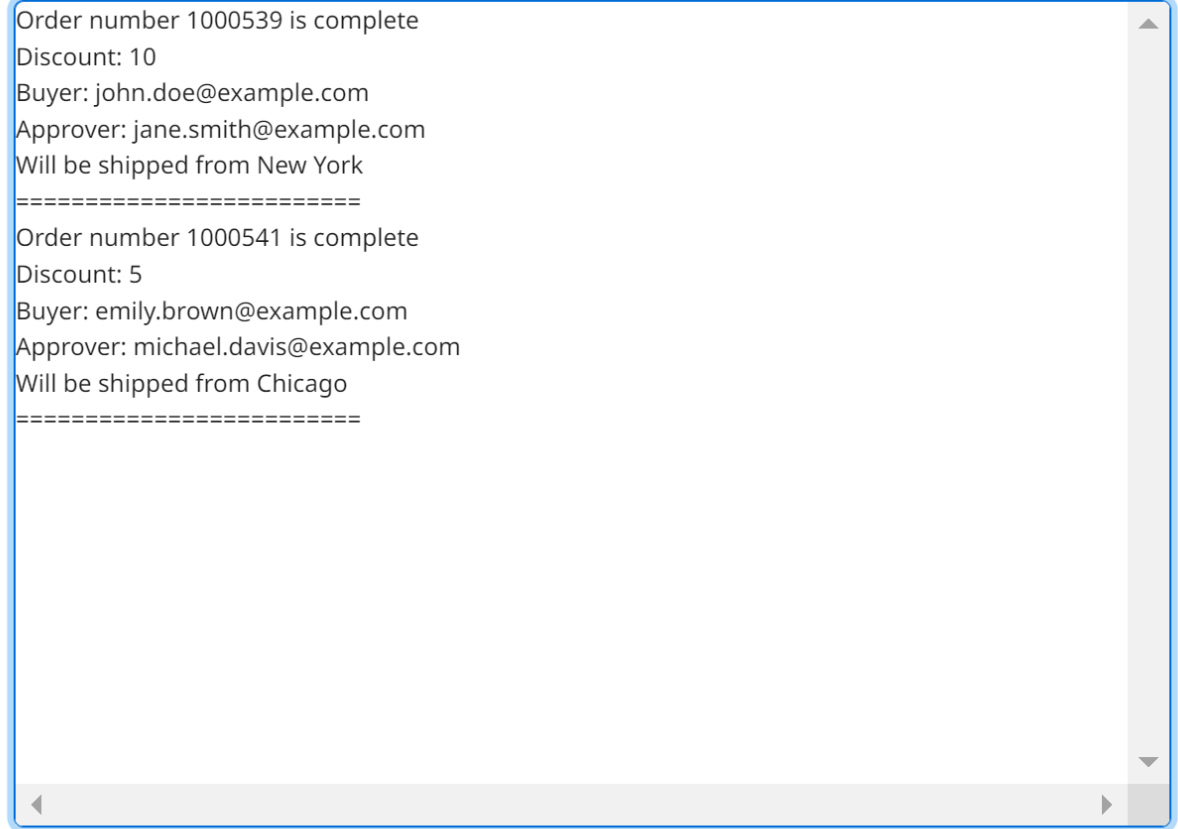


6. **Save** and **Test** the process. Review the output of both the **Accepted** and **Rejected** paths and compare this to the email(s) you received.



7. The output for the **Accepted** path will look like the image below:

Document Viewer



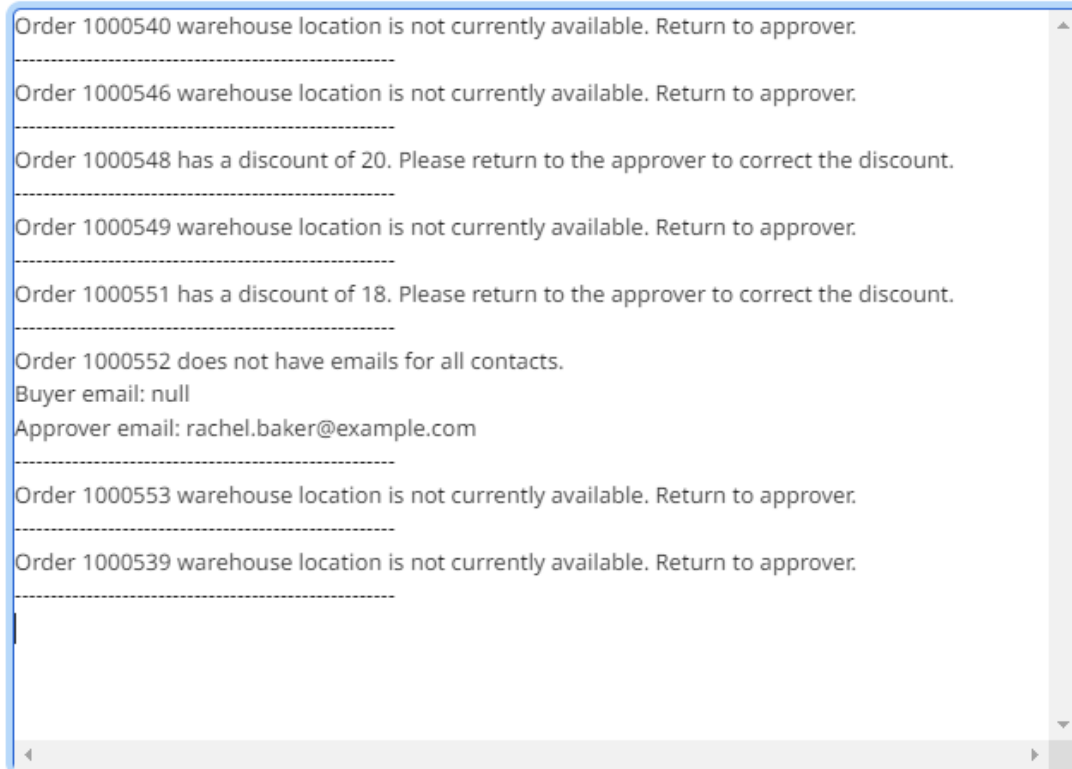
[Download Original Document](#)

Size 330 bytes

Close Document Viewer

8. The output for the **Rejected** path will look like the image below:

Document Viewer



[Download Original Document](#)

Size 1131 bytes

[Close Document Viewer](#)

Activities Wrap-Up

Congratulations, you completed testing your **Business Rules step** solution and have completed all activities.



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